Complex numbers

1 A complex number

2 $0 + bi$ or bi

3

a $\sqrt{-1}$ **b** -1

4

a 0 **b** $-\sqrt{10}$

c Nonreal complex/pure imaginary

5

a $\sqrt{5}i$ **b** $\sqrt{14}$

Operations with complex numbers

6 $9 + 2i$

7 No, $-4i$

8 No, $\dfrac{6 \pm 36i}{2} = 3 \pm 18i$

9

a $-10i$ **b** $3i\sqrt{2}$ **c** $9 - 8i$ **d** $2 + 5i$

10 -8

11

a -19 **b** 5 **c** $\sqrt{5}$ **d** $\dfrac{1}{3}$

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12

a $10 + 9i$ **b** $3 + 9i$ **c** $-3 + 5i$ **d** $6 - 11i$

e $-3.3 + 2.6i$ **f** $-11 + 16i$ **g** $-11 - 5i$ **h** $12 - 6i$

i $\frac{7}{6} - \frac{i}{20}$

13

a $-18 + 30i$ **b** $14 + 21i$ **c** $-10 + 8\sqrt{10}i$ **d** $96 - 16i$

e 65 **f** -29 **g** $-21 - 20i$ **h** $-\frac{1}{4} + \frac{3}{4}i$

i $16i + 12$ **j** $-14i - 48$

14

a

i -1 **ii** 1 **iii** -1 **iv** 1

b

i Yes **ii** No **iii** Yes **iv** No

15

a $-2i$ **b** $-v$ **c** -4 **d** 10

e $-27i$ **f** 16 **g** $-13 - 7i$ **h** $-7 - 2i$

Complex conjugates

16 Complex conjugates

17 A real number

18

a 20 **b** 39 **c** 26 **d** 25

e 17 **f** 3 **g** 21 **h** 5

19

a $2 - 3i$ **b** $-5i + 2$ **c** $\frac{5i}{6}$ **d** $2 + 5i$

e $5 + i$ **f** $2 - i$ **g** $3 + 2i$ **h** $-\frac{5}{13} - \frac{12i}{13}$

i $-\frac{2}{17} - \frac{9i}{17}$ **j** $\frac{2}{5} + \frac{6i}{5}$

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20

a 10

b 31

21

a $\dfrac{67-47i}{34}$

b $-7-10i$

22 $\dfrac{16}{5}-5i$

Application

23 $33+50i$

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